Some important information regarding exam two-draft 1

Exam two: Thursday, Feb. 20th
Extra Office Hours: I will be holding extra office hours on Saturday, Sunday, TBA
Bring UCLA photo ID.

Topics covered on exam (Second exam is NOT cumulative)

1. Asset Approach and the Bond Market: ***Relevant study questions****: see SQ#2: 1 -3.* What is the Asset Approach? Know the various shift variables for the bond S&D curves (and its counterpart in the LF market) and why they shift a curve. The questions here are really no more than S&D questions. These questions will be standard S&D questions. Keys: shift variables for bond supply and bond demand D_{bonds} (wealth, expected future return on bond relative to other assets, relative risk on bond, relative liquidity of bond); S_{bonds} (EFPI, government deficit, expected inflation). NOTE: As I mentioned in class since exam one already covered the Fisher effect, I will not be assuming a change in expected inflation for these questions)

2. Term of structure of interest rates. *Relevant study questions: SQ#2, 1-5, under heading "Term Structure of interest rates"*. Important items included expectations theory (what is this theory, what is the logic of this theory—think S&D story, relevant equations, what can we deduce from this theory). Liquidity premium theory (same points as under expectations theory). What is interest rate risk? Relationship between these theories and the yield curve. Using Yield curve to forecast recessions. How to read Fed charts with ($i_{10\text{ year}} - i_{3\text{-month}}$) gap.
Some important equations you should know for this exam:
1. $(1+ i_{02})^2 = (1+i_{01})(1+ i_{11})$ for Expectations Theory. You also need to know the approximate version.
2. $(1+ i_{02})^2 = (1+i_{01})(1+ i_{11}) + \eta$ for Liquidity Premium Theory. Also know the approximate version.
3. n year version of the above.

3. Money supply and the Fed AND Tools of monetary management, *Relevant study questions: see SQ#2, 1-9 under heading "THE FED AND MONEY SUPPLY"*. Fed and institutional details given in class. Key assets and liabilities on bank and Fed balance sheets. What are reserves? Simplified banking system (be able to do T-account story). What is the monetary base and money multiplier in advance money supply model? What factors influence the money multiplier, who controls them, how do they impact M1? Specially, know the factors that influence c (bank panics, level of illegal activity, rates on alternative assets, payments technology, foreign demand for currency) and e (difference between federal funds rate and rate paid on reserves, bank expectations about future withdrawals, bank expectations about future riskiness of assets), and s (alternative rates). How can Fed take offsetting actions? ~~What is meant by borrowed reserves and non-borrowed reserves? What are the three conventional tools of monetary management (OMO, Discount rate, rr) and how do they each work?~~ Be able to look at data on, for example, the MB,M1, mm, etc. and to analyze/explain these data in the context of the money supply model (equation). Short discussion on MV = PY
Some important equations you should know for this exam:
$\Delta R * (1/rr) = \Delta D$ (simplified banking system) and
$M1 = MB * (1+c)/(c+ rr +e)$ (Advanced banking system model—old
$M1 = MB *(1+c+s)/(c+ e)$ (Advanced banking system model—new, assuming rr = 0)
$MB = C+R$, $M1 = C +D$ (OLD), $M1 = C + D + S$ (NEW-post May 2020)
$c = C/D$ (currency ratio), $e = ER/D$ (excess reserve ratio) , $s = S/D$ (savings ratio)

ER = R – rr*D (tells us how to calculate excess reserves in simplified banking system model),
~~$MB = MB_n + BR$ (MB_n is also called NBR, non-borrowed reserves)~~

Also do the practice exam questions. ONLY those questions that relate to the above topics.

NOT ON EXAM TWO—market for reserves (R^s,R^d stuff etc.) and the topics we have covered after that.

SAMPLE EXAM questions

. **Test A**

1. Suppose the yield curve has a downward slope. Under the Liquidity Premium Theory, what can we conclude about the public's interest rate expectations in a two-period model?

a. $i_{01} > i_{11}$.
b. $i_{01} < i_{11}$.
c. $i_{01} = i_{11}$.
d. either $i_{01} < i_{11}$ or $i_{01} = i_{11}$ is possible, it depends on the size of the liquidity premium.
e. either $i_{01} > i_{11}$ or $i_{01} = i_{11}$ is possible, it depends on the size of the liquidity premium.
f. either $i_{01} > i_{11}$ or $i_{01} < i_{11}$ or $i_{01} = i_{11}$ is possible, it depends on the size of the liquidity premium.

2. Assume the current yield on a two-year bond (two-year loan) is i_{02}= 5% and the current yield on a one-year bond (one-year loan) is i_{01} = 4%. If the yield expected on an one-year bond purchased one year from today is i_{11}= 6%, then would this set of interest rates be considered an equilibrium according to the liquidity premium theory? If so, explain why? If not, explain why?

a. Would not be considered an equilibrium under the LP theory. The purchase of consecutive one-year bonds would be preferred to the purchase one two-year bond. The ensuing adjustments would then cause the price of one year bonds to rise and the price of two-year bonds to fall.
b. Would not be considered an equilibrium under the LP theory. The purchase of one two-year bond would be preferred to the purchase consecutive two one-year bonds. The ensuing adjustments would then cause the price of one year bonds to fall and the price of two-year bonds to rise.
c. Would not be considered an equilibrium under the LP theory. The purchase of consecutive one-year bonds would be preferred to the purchase one two-year bond. The ensuing adjustments would then cause the price of one year bonds to fall and the price of two-year bonds to rise.
d. Would not be considered an equilibrium under the LP theory. The purchase of one two-year bond would be preferred to the purchase consecutive two one-year bonds. The ensuing adjustments would then cause the price of one year bonds to rise and the price of two-year bonds to fall.
e. It would be considered an equilibrium because the annual return on the purchase of one two-year bond equals the average annual return on the purchase of two consecutive one-year bonds. There would be no additional change in bond prices.
f. It would be considered an equilibrium because the annual return on the purchase of one two-year bond equals the average annual return on the purchase of two consecutive one-year bonds. The demand for one year bonds would change until the current period one-year yield was also 5%.

3. You have been asked to provide a forecast of the inflation rate for *next* year. You have good reason to believe that the real rate of interest will be constant over the next few years. Suppose you observe a horizontal yield curve. From this you could to determine:

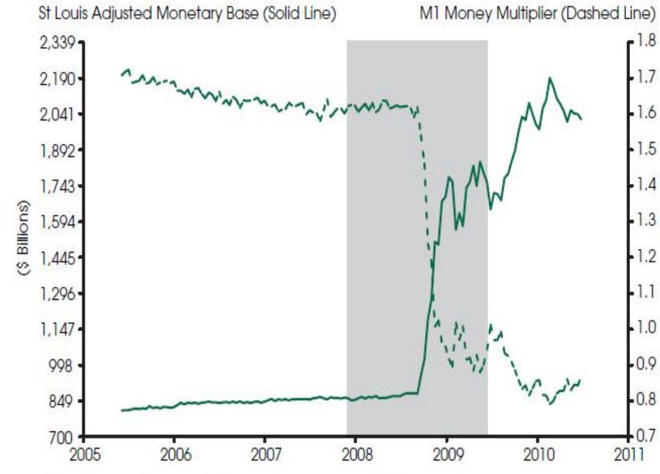
a. by the Fisher equation and the implied forward rate (i_{11}) the inflation rate next year is expected to go down, assuming the Expectations Theory is correct.
b. by the Fisher equation and the implied forward rate (i_{11}) that the inflation rate next year is expected to go down, assuming the Liquidity Premium Theory is correct.
c. by the Fisher equation and the implied forward rate (i_{11}) it is uncertain whether the inflation rate next year will go or down, assuming the Expectations Theory is correct.
d. by the Fisher equation and the implied forward rate (i_{11}) it is uncertain whether the inflation rate next year will go or down, assuming the Liquidity Premium Theory is correct.
e. by the Fisher equation and the implied forward rate (i_{11}) the inflation rate next year is expected to go up, assuming the Expectations Theory is correct.
f. it is not possible to determine if the inflation rate will go up, down, or remain the same regardless of which theory is correct.

4. Use the following information to answer the next three questions. Assume a simplified banking system (that is, use the same assumptions we made in class when we discussed the simplified banking system). Suppose that the Fed purchases $100,000 of bonds directly from Bank A. The required reserve ratio (rr) for demand deposits is 20%. Consistent with our money expansion story, what is the maximum loan <u>Bk. **C**</u> can make ?
[Hint: you should not have to go through the entire T-account story to answer this]
a. $100,000.
b. $20,000.
c. $80,000.
d. $16,000.
e. $64,000.
f. $12,800.
g. $51,200.
h. $40,960.
i. none of the above.

5. Using the information from the previous question. The ultimate increase in LOANS for the <u>entire</u> banking system in final equilibrium is:
a. $100,000.
b. $500,000.
c. $800,000.
d. $400,000.
e. $180,000.
f. $409,600.
g. $320,000.
h. $640,000.
i. none of the above.

6. Assume the banking system described above is in equilibrium following the purchase of bonds by the Fed as indicated in the previous questions. The Fed however has now decided that it prefers a money supply <u>total</u> of $800,000. Before taking any actions on this and for independent reasons, the Fed has also decided to decrease the required reserve ratio to 10%. It order to achieve its target money supply it should engage in:
a. an open market purchase of bonds of $300,000.
b. an open market sale of bonds of $300,000.
c. an open market purchase of bonds of $60,000.
d. an open market sale of bonds of $60,000.
e. an open market purchase of bonds of $30,000.
f. an open market sale of bonds of $30,000.
g. an open market sale of bonds of $20,000.
h. an open market purchase of bonds of $20,000.
i. none of the above.

7. See chart below. The monetary base (MB) is measured on the left vertical (solid line) and the money multiplier is measured on the right vertical axis (dashed line).



NOTE: The shaded area indicates the most recent U.S. recession.

You have been asked to provide POSSIBLE explanations for the patterns depicted in this chart since 2008 and for the likely impact of these changes on M1. Which of the following sets of explanations provide possible reasons for these changes? (Assume that the relevant multiplier value is the one listed in 2008)

a. Since 2008, the Fed has dramatically increased non-borrowed reserves; since 2008 there has been a dramatic decline the excess reserve ratio (e); the likely impact is a large increase in M1 matching the large increase in the monetary base.
b. Since 2008, the Fed has dramatically decreased non-borrowed reserves; since 2008 there has been a dramatic decline the excess reserve ratio (e); the likely impact is very little change in M1 despite the large increase in the monetary base.
c. Since 2008, the Fed has dramatically increased non-borrowed reserves; since 2008 there has been a dramatic increase in the currency ratio (c); the likely impact is a large increase in M1 matching the large increase in the monetary base.
d. Since 2008, the Fed has dramatically decreased non-borrowed reserves; since 2008 there has been a dramatic increase in the currency ratio (c); the likely impact is a large increase in M1 matching the large increase in the monetary base.
e. Since 2008, the Fed has dramatically increased non-borrowed reserves; since 2008 there has been a dramatic increase the excess reserve ratio (e); the likely impact is a relatively small increase in M1 despite the large increase in the monetary base.
f. Since 2008, the Fed has cut the required reserve ratio and kept non-borrowed and borrowed reserves relatively constant; the likely impact on the money supply is a large increase matching the increase in the monetary base.

15. Assume the money multiplier is greater than one. There is an increase in excess reserve ratio that the Fed did not expect. At the same time the currency ratio decreased. Which of the following statements is correct? (assume federal funds rate between the discount rate and rate on reserves)

a. To maintain a fixed money supply the Fed should engage in an open market sale of bonds.
b. To maintain a fixed money supply the Fed should raise the discount rate.
c. To maintain a fixed money supply the Fed should engage in an open market purchase of bonds.
d. To maintain a fixed money supply the Fed should lower the rate on reserves.
e. two of the above would accomplish the goal of maintaining a fixed money supply.
f. it is uncertain what action the Fed should take.

17. See chart below. This chart shows the non-bank public's currency ratio (c) with it being around 30% in the early 1990s.

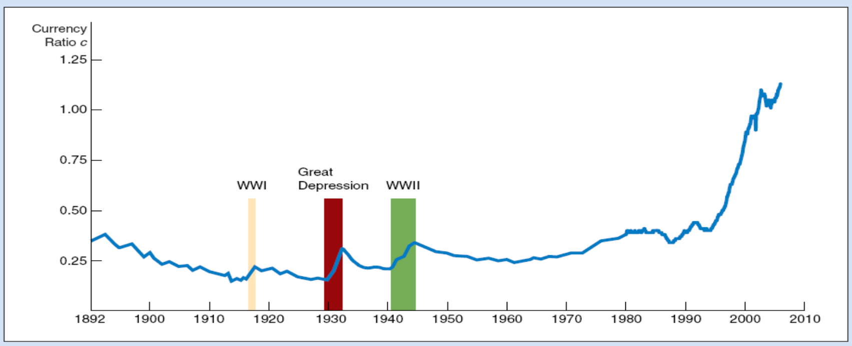


FIGURE 1 **Currency-Checkable Deposits Ratio: 1892–2005**
Sources: Federal Reserve *Bulletin* and *Banking and Monetary Statistics*. www.federalreserve.gov/releases/h6/hist/h6hist1.txt

Since the early 1990s there has been a substantial increase in this ratio due to the introduction of ATM's lower cost of acquiring currency. For most of this period the money multiplier has exceeded one (assume for this question it is greater than one for the entire period in question). Prior to 2008 the excess reserve ratio (e) was close to zero, so for simplicity we will assume it was zero. The required reserve ratio was, let us assume, constant at 10%. Over this same period the money supply M1 has increased but not substantially. From the above data, we can determine the money multiplier was approximately _________ in the early 1990s (you can leave your answer in ratio form, but you should be able to approximately see what it is equal to) and by 2007 is approximately ___________ (again, you can leave your answer in ratio form).
If the money supply was gently rising over this period, we can deduce that the monetary base approximately changed by what amount ______________________ (for this blank, looking for answer such as "roughly stayed the same", "roughly doubled", "roughly tripled," etc.

Term structure of interest rates

1. a. What is the Expectations Theory (assume two periods)? In answering this question I am simply looking for what key assumption it makes and what is implies in terms of equations. Neither answers involves much writing, so be brief. (6 pts.)

1.b. Explain the logic of this theory by indicating what happens when there is *not* an equilibrium set of interest rates. Use the appropriate graphs. Again, the answer does not involve much writing. (6 pts.)

SAMPLE EXAM-answers

Multiple choice. Select the best option. Each MC/Fill-in-the blank is worth 2.5 pts. Test A

1. Suppose the yield curve has a downward slope. Under the Liquidity Premium Theory, what can we conclude about the public's interest rate expectations in a two-period model?

a. $i_{01} > i_{11}$.

$i_{02} > [i_{01} + i_{11}]/2$ + risk premium (+) → i_{02} =8% (for example) minus (+) = $[9 + i_{11}]/2$ ---->i01 > i11.

b. $i_{01} < i_{11}$.

c. $i_{01} = i_{11}$.

d. either $i_{01} < i_{11}$ or $i_{01} = i_{11}$ is possible, it depends on the size of the liquidity premium.

e. either $i_{01} > i_{11}$ or $i_{01} = i_{11}$ is possible, it depends on the size of the liquidity premium.

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2. Assume the current yield on a two-year bond (two-year loan) is i_{02}= 5% and the current yield on a one-year bond (one-year loan) is i_{01} = 4%. If the yield expected on an one-year bond purchased one year from today is i_{11}= 6%, then would this set of interest rates be considered an equilibrium according to the liquidity premium theory? If so, explain why? If not, explain why?

5% vs. [4+6]/2 → cannot be equilibrium because not indifferent between options if each pays an average annual rate of 5%; would rather hold the short-term options, thus would increase demand for one-year bonds (price up, yield down) and decrease purchase of 2 year bond (price down, yield up)

a. Would not be considered an equilibrium under the LP theory. The purchase of consecutive one-year bonds would be preferred to the purchase one two-year bond. The ensuing adjustments would then cause the price of one year bonds to rise and the price of two-year bonds to fall.

b. Would not be considered an equilibrium under the LP theory. The purchase of one two-year bond would be preferred to the purchase consecutive two one-year bonds. The ensuing adjustments would then cause the price of one year bonds to fall and the price of two-year bonds to rise.

c. Would not be considered an equilibrium under the LP theory. The purchase of consecutive one-year bonds would be preferred to the purchase one two-year bond. The ensuing adjustments would then cause the price of one year bonds to fall and the price of two-year bonds to rise.

d. Would not be considered an equilibrium under the LP theory. The purchase of one two-year bond would be preferred to the purchase consecutive two one-year bonds. The ensuing adjustments would then cause the price of one year bonds to rise and the price of two-year bonds to fall.

e. It would be considered an equilibrium because the annual return on the purchase of one two-year bond equals the average annual return on the purchase of two consecutive one-year bonds. There would be no additional change in bond prices.

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3. You have been asked to provide a forecast of the inflation rate for *next* year. You have good reason to believe that the real rate of interest will be constant over the next few years. Suppose you observe a horizontal yield curve. From this you could to determine:

a. by the Fisher equation and the implied forward rate (i_{11}) the inflation rate next year is expected to go down, assuming the Expectations Theory is correct.
b. by the Fisher equation and the implied forward rate (i_{11}) that the inflation rate next year is expected to go down, assuming the Liquidity Premium Theory is correct. [under LP, this would mean i11 < i01, would a constant real rate this would mean expect inflation to fall]
c. by the Fisher equation and the implied forward rate (i_{11}) it is uncertain whether the inflation rate next year will go or down, assuming the Expectations Theory is correct.
d. by the Fisher equation and the implied forward rate (i_{11}) it is uncertain whether the inflation rate next year will go or down, assuming the Liquidity Premium Theory is correct.
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4. Use the following information to answer the next three questions. Assume a simplified banking system (that is, use the same assumptions we made in class when we discussed the simplified banking system). Suppose that the Fed purchases $100,000 of bonds directly from Bank A. The required reserve ratio (rr) for demand deposits is 20%. Consistent with our money expansion story, what is the maximum loan Bk. **C** can make ? $100{,}000(1\text{-rr})^3 = 100\text{k}(.80)^3 = 51.2\text{K}$
[Hint: you should not have to go through the entire T-account story to answer this]
a. $100,000.
b. $20,000.
c. $80,000.
d. $16,000.
e. $64,000.
f. $12,800.
g. $51,200.
h. $40,960.
i. none of the above.

5. Using the information from the previous question. The ultimate increase in LOANS for the entire banking system in final equilibrium is:
First find D -→ D = 100,000(1/rr) = 500,000. For LOANS, substract from 500K the reserves of 100k.
a. $100,000.
b. $500,000.
c. $800,000.
d. $400,000.
e. $180,000.
f. $409,600.
g. $320,000.
h. $640,000.
i. none of the above.

6. Assume the banking system described above is in equilibrium following the purchase of bonds by the Fed as indicated in the previous questions. The Fed however has now decided that it prefers a money supply total of $800,000. Before taking any actions on this and for independent reasons, the Fed has also decided to decrease the required reserve ratio to 10%. It order to achieve its target money supply it should engage in:

→ D = 100,000(1/rr) = 500,000 but now 800,000= Reserves (1/.10) → Reserves = 80,000. This would require a decrease in reserves of 20,000 or an O.M. Sale of 20k.

a. an open market purchase of bonds of $300,000.
b. an open market sale of bonds of $300,000.
c. an open market purchase of bonds of $60,000.
d. an open market sale of bonds of $60,000.
e. an open market purchase of bonds of $30,000.
f. an open market sale of bonds of $30,000.
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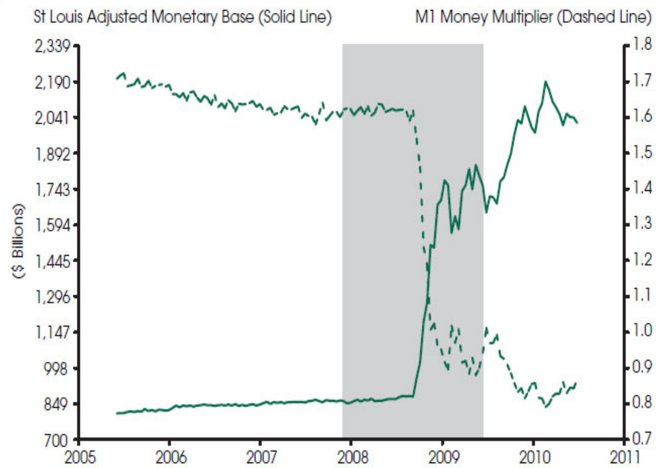

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c. Since 2008, the Fed has dramatically increased non-borrowed reserves; since 2008 there has been a dramatic increase in the currency ratio (c); the likely impact is a large increase in M1 matching the large increase in the monetary base.
d. Since 2008, the Fed has dramatically decreased non-borrowed reserves; since 2008 there has been a dramatic increase in the currency ratio (c); the likely impact is a large increase in M1 matching the large increase in the monetary base.
e. Since 2008, the Fed has dramatically increased non-borrowed reserves (this explains the rise in MB); since 2008 there has been a dramatic increase the excess reserve ratio (e) (this explains the fall in mm) the

15. Assume the money multiplier is greater than one. There is an increase in excess reserve ratio that the Fed did not expect. At the same time the currency ratio decreased. Which of the following statements is correct? (assume federal funds rate between the discount rate and rate on reserves)
By itself these rise in e lowers mm, and lower c raises mm given mm>1. This uncertain impact on mm and M1
a. To maintain a fixed money supply the Fed should engage in an open market sale of bonds.
b. To maintain a fixed money supply the Fed should raise the discount rate.
c. To maintain a fixed money supply the Fed should engage in an open market purchase of bonds.
d. To maintain a fixed money supply the Fed should lower the rate on reserves.
e. two of the above would accomplish the goal of maintaining a fixed money supply.
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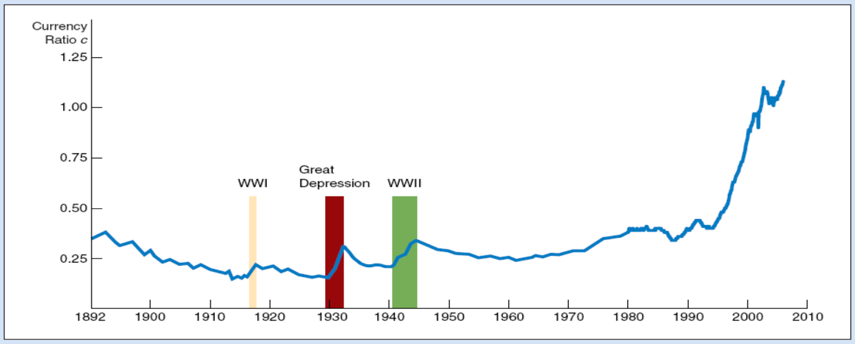

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Since the early 1990s there has been a substantial increase in this ratio due to the introduction of ATM's lower cost of acquiring currency. For most of this period the money multiplier has exceeded one (assume for this question it is greater than one for the entire period in question). So by itself, rise in c lowers mm and M1/.Prior to 2008 the excess reserve ratio (e) was close to zero, so for simplicity we will assume it was zero. The required reserve ratio was, let us assume, constant at 10%. Over this same period the money supply M1 has increased but not substantially. From the above data, we can determine the money multiplier was approximately mm = 1+.4 / (.4+.10+0)= 1.4/.5 or roughly 2.8_ in the early 1990s (you can leave your answer in ratio form, but you should be able to approximately see what it is equal to) and by 2007 is approximately mm=1+1/1+.10+0=2/1.1=1.8 or 1.9_ (again, you can leave your answer in ratio form).
If the money supply was gently rising over this period, we can deduce that the monetary base approximately changed by what amount _roughly doubled__ (for this blank, looking for answer such as "roughly stayed the same", "roughly doubled", "roughly tripled," etc.

Term structure of interest rates **test A**

1. a. What is the Expectations Theory (assume two periods)? In answering this question I am simply looking for what key assumption it makes and what is implies in terms of equations. Neither answers involves much writing, so be brief. (6 pts.)

1.b. Explain the logic of this theory by indicating what happens when there is *not* an equilibrium set of interest rates. Use the appropriate graphs. Again, the answer does not involve much writing. (6 pts.)

1a. option A (buying of one 2 year bond) is perfect substitute for option B (buying two consecutive 1 year bonds of similar default risk). So i02 = (i01 + i11) / 2 in equilibrium

1b. Suppose i02 > (i01 + i11) / 2 then better to choose option A. Therefore demand for 2 yr bonds rises (driving up its price and lowering its yield) and demand for one year bonds falls (lowering prices, raising yields)

You put in S&D graphs.